

Operators



"Every problem you solve today is one step closer to your dream company."

OPERATORS

Operators are symbols used to perform operations on data (variables/values).

Operand

An operand is the value or variable on which the operator works.

Operator

An operator is a symbol that performs an operation on values.



$\boxed{2} + \boxed{4}$ } operands

$\boxed{20} * \boxed{40}$

operator

TYPES OF OPERATOR

Unary Operator

Which requires only one operand.

► $++x, y--$

Binary Operator

Which requires two operands.

► $x + y$



Unary (1) $++x, x--, --x, x++$

Binary (2) $x+y, x/y, x<y, x>y$

Ternary (3) $x ? y : z$

Arithmetic Operator

Used for mathematical calculations

+ (Addition)

- (Subtraction)

*** (Multiplication)**

/ (Division)

% (Modulus - remainder)



Arithmetic Operator Precedence

Associativity of Arithmetic Operator is from Left to Right.

Below is the priority of operators.

High

*** (Multiplication)**

/ (Division)

% (Modulus - remainder)

Low

+ (Addition)

- (Subtraction)

$$x = 10 + 9 \cdot 3 / 1 * 6$$

8,

left to Right

100

9, 10

Logical Operator $\rightarrow$ True | False

Used to combine multiple conditions

Mainly returns true or false.

&& (AND)

|| (OR)

! (NOT)



Logical Operator Precedence

Associativity of logical Operator is from Left to Right.

Below is the priority of operators.

High

!(NOT) 1

&&(AND) 2

Low

|| OR 2

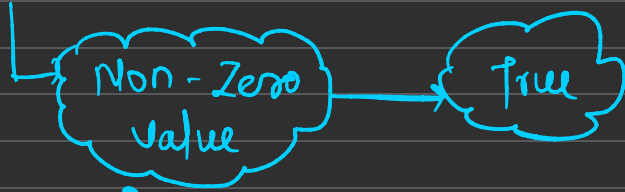
NOT (!)

Alter the condition

$\text{True}! = \text{False}$

$\text{False}! = \text{True}$

$7! \rightarrow \text{True}! \rightarrow \text{False}$



$-4! \Rightarrow \text{True}! \rightarrow \text{False}$

$0! \rightarrow \text{False}! \rightarrow \text{True}$

$0.25!$ = $\text{True}!$ = False

AND ($\&\&$) (Multiplication)

x	y	$(x \& y)$ output
0	0	0
0	1	0
1	0	0
1	1	1

$x \&\& y$

→ if x is True,
then y is checked

if x is False,
 y is not checked

OR (||) Addition

x	y	(x+y)
0	0	0
0	1	1
1	0	1
1	1	1

x || y

↳ if x is true,

y is not checked

if x is False,

y will be checked

Logical && (AND) Operator

X && Y

If X is true then and then only Y is checked.

If X is false, Y will not be checked.

Logical || (OR) Operator

X || Y

If X is true then and then Y is not checked.

If X is false, then Y is checked.



Relational Operator

Conditional



Used to compare two values

Mainly returns true or false.

== (equal to)

!= (not equal)

> (greater than)

< (less than)

>=, <=



$$x = 10 \quad y = \underline{\underline{100}}$$

$$x == y \rightarrow \text{True}$$
$$10 == 10 \rightarrow$$

$$x == y \rightarrow \text{False}$$
$$10 == 100$$

$$x < y \rightarrow \text{True}$$
$$10 < 100$$

$$x > y \rightarrow \text{False}$$
$$10 > 100$$

$$m = 5, n = 5$$

$$m < n = \text{False}$$

$$m \geq n = 5 \geq 5$$

$$m \leq n$$
$$5 \leq 5$$

(True)

$$5 < 5 \times$$
$$\underline{\underline{5 == 5}}$$

Relational Operator Precedence

Associativity of relational Operator is from Left to Right.

Below is the priority of operators.

High

`!=` (not equal)

`>` (greater than)

`<` (less than)

Low

`>=`, `<=`

`==` (equal to)

Increment Decreament

Mainly contains pre-increment and post-increment.

Mainly contains pre-decrement and post-decrement.

`++X`

`X++`

`--X`

`X--`



Increment

1.7 pre-increment ($++x$)

↳ first increase, then you will use

```
int x = 10
y = ++x
print(x, y)
```

~~int x = 10~~
→ ~~x = x + 1~~
~~y = x~~
~~print(x, y)~~

x	y
10	11
11	

10 11

11 10

11 11

— post-increment

↓ first use it, then increase

```
int x = 10
y = x++
print(x, y)
```

→ ✓ int x = 10
✓ y = x
✓ x = x + 1
print(x, y)

x	y
11	10

11 10

int m = 10

int x, y

~~x = ++m~~

~~y = m++~~

x--

--y

x = x - y

print (x)

~~int m = 10~~

~~int x, y~~

~~m = m + 1~~

~~x = m~~

~~y = m~~

~~m = m + 1~~

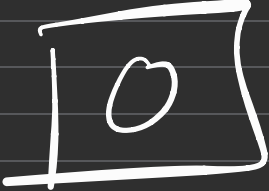
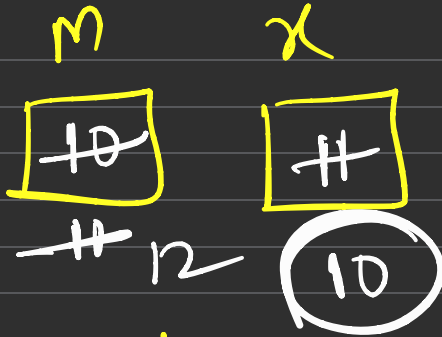
~~x = x~~

~~x = x + 1~~

~~y = y + 1~~

~~y = y~~

~~x = x - y~~

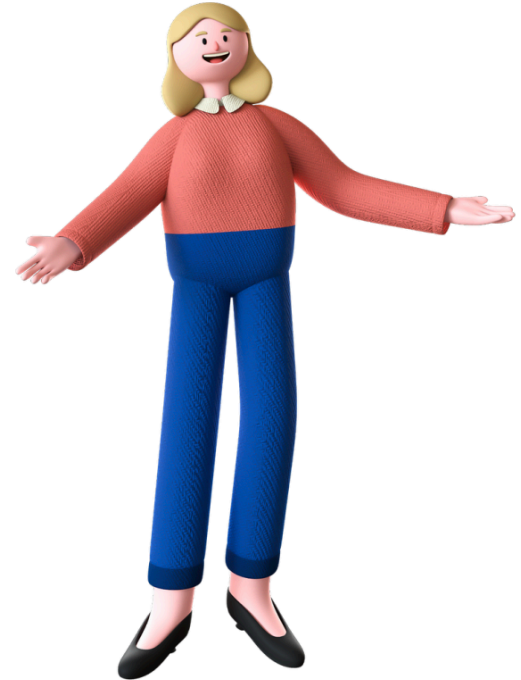


Pre-Increment = first increase and then use it.

Post-Increment = first use it and then increase.

Pre-decrement = first decrease and then use it.

Post-decrement = first use it and then decrease.



int x = 3 > 2 < 1 > 4 < 6 == 0

> <

1 < 1 > 4 < 6 == 0

==

0 > 4 < 6 == 0

< >

0 < 6 == 0

!=

1 == 0

0 = False

int $x = \underline{30 > 20} > 0 \mid = 2 < 50 > 40 \mid = 50$

$1 > 0 \mid = 2 < 50 > 40 \mid = 50$

$1 \mid = 2 < 50 > 40 \mid = 50$

free

$1 < 50 > 40 \mid = 50$

$1 > 40 \mid = 50$

$0 \mid = 50$

0000a
0010
0210

int a=1, b=-1, c=0, d

d = --a || ++b && c++
printf (a, b, c, d)

x || y

++b && c++
0 && 0

a b
[0] [~~0~~]

c d
[0] [0]

int a=1, b=-1, c=0, d

$$d = \underbrace{--a}_{1} \parallel \underbrace{(b++ \&\& c++)}_{11 \quad 4}$$

0 -1 0 0

0 0 1 0

a	b
1 0	-1 0
c	d
0 1	0

int a=2, b=2, c=0, d=2, m

m = (a++ ~~||~~ b++) ~~||~~ (c++ ~~||~~ d++)

True

c++ b b d++

a	b	c	d	m
2	2	0	2	
3	3	1	2	0

int x = 10

int y = sizeof(x)

printf(x, y)

int x = 10

int y = sizeof(x = x + 1)

printf(x, y)

10 2/4

10 4